

Airframe Design - Optimization: Part III

Research Skills: Version Control

As part of this assignment, you will be expected to further develop introductory skills in version control. In order to do so, complete the following:

- a. Create at least 3 github issues (which should be based on the feedback you received in the previous assignment).
- b. Create a branch for this assignment.
- c. Add, Commit, and Push your assignment submission to your github repository.
- d. Open a pull request as a part of your submission for this assignment.

Technical Tasks: Improved Optimization

Based on your experience completing the previous assignment, in combination with the feedback you received, you should have identified several areas in which you might improve your optimization. The overall goal for the technical tasks of this assignment are to make improvements according to that experience/feedback.

- a. With your graduate student mentor, identify at least one area in which you can improve your optimization. Incorporate this into your optimization. Some ideas include optimizing the twist and chord distributions simultaneously, using splines to define you chord distribution, or using algorithmic differentiation.

Writing Tasks: Methodology and Results

As you now have a better handle on what you're doing, it would be best to flesh out the details of your methodology as well as formally outline the results section of your final report draft. For this assignment, complete the following writing tasks:

- a. Complete a full draft of your methodology section, including all of the information required to reproduce your work.
- b. Outline the results section of your report, including placeholders for figures and cues for discussion content to include.